Physics P2: Solutions

Motion, Forces, and Conservation · MAIS PAC

Problem 1.

(a) Conservation of mechanical energy between launch (height $\mathbf{h}$, speed $\mathbf{v}_0$) and impact (height 0, speed $\mathbf{v_f}$) gives $(1)/(2)\mathbf{m}\mathbf{v}_0^2 + \mathbf{mgh} = (1)/(2)\mathbf{m}\mathbf{v_f}^2$, so

$$\mathbf{v}_1 = \mathbf{v}_2 = \mathbf{v}_3 = \sqrt{(\mathbf{v}_0^2 + 2\mathbf{gh})}. \tag{1}$$

All three stones strike the ground at the **exact same speed** (though at different angles).

(b) Stone 1 rises from cliff height $\mathbf{h}$ and returns to cliff height $\mathbf{h}$ after time $\Delta\mathbf{t} = 2\mathbf{v}_0 \sin\alpha / \mathbf{g}$, at which moment its downward vertical velocity is $-\mathbf{v}_0 \sin\alpha$ and its horizontal velocity is $\mathbf{v}_0 \cos\alpha$—**identical** to the initial state of stone 3 at the moment stone 3 was thrown! From that point onward, stone 1 traces the exact same descent to the ground as stone 3. Hence

$$\mathbf{t}_1 - \mathbf{t}_3 = (2\mathbf{v}_0 \sin\alpha)/(\mathbf{g}), \tag{2}$$

which depends only on the extra loop stone 1 makes above the cliff, not on $\mathbf{h}$ at all.

Problem 2.

(a) At the top of the loop (height $2\mathbf{R}$), gravity $\mathbf{mg}$ and inward track normal force $\mathbf{N}_{top} \geq 0$ both point downward toward the center: $\mathbf{mg} + \mathbf{N}_{top} = \mathbf{m}\mathbf{v}_{top}^2/\mathbf{R}$. At the threshold of losing contact ($\mathbf{N}_{top} = 0$), $\mathbf{v}_{top}^2 = \mathbf{gR}$. Conservation of energy from rest at height $\mathbf{h}_{min}$ gives

$$\mathbf{mgh}_{min} = \mathbf{mg}(2\mathbf{R}) + (1)/(2)\mathbf{m}(\mathbf{gR}) \Rightarrow \mathbf{h}_{min} = (5)/(2)\mathbf{R}. \tag{3}$$

(b) At height $\mathbf{R}$ (with $\mathbf{h} = (5)/(2)\mathbf{R}$), energy conservation gives $(1)/(2)\mathbf{m}\mathbf{v}^2 = \mathbf{mg}((5)/(2)\mathbf{R} - \mathbf{R}) = (3)/(2)\mathbf{mgR}$, so $\mathbf{v}^2 = 3\mathbf{gR}$. Therefore the inward horizontal centripetal acceleration is $\mathbf{a_r} = \mathbf{v}^2/\mathbf{R} = 3\mathbf{g}$, while the track wall is vertical so only gravity acts tangentially, giving downward tangential acceleration $\mathbf{a_t} = \mathbf{g}$. Since $\mathbf{a_r}$ and $\mathbf{a_t}$ are perpendicular,

$$|\mathbf{a}| = \sqrt{((3\mathbf{g})^2 + \mathbf{g}^2)} = \sqrt{10}\,\mathbf{g} \approx 3.16\mathbf{g}, \tag{4}$$

directed inward toward the loop's axis and downward at angle $\arctan(1/3) \approx 18.4°$ below the horizontal.

Problem 3.

(a) Both balls reach the floor with speed $\mathbf{v}_0 = \sqrt{(2\mathbf{gh})}$. Right after the basketball rebounds elastically off the floor, its velocity is $+\mathbf{v}_0$ (upward) while the tennis ball is still moving at $-\mathbf{v}_0$

(downward). Switch into the inertial frame moving upward with the heavy basketball ($+\mathbf{v}_0$): in this frame the basketball is a massive stationary wall (since $\mathbf{M} \gg \mathbf{m}$), and the tennis ball approaches it downward at relative speed $2\mathbf{v}_0$. An elastic bounce reverses its relative velocity to $+2\mathbf{v}_0$ upward in the basketball's frame. Transforming back to the room frame gives

$$\mathbf{v}_{\text{tennis}} = \mathbf{v}_0 + 2\mathbf{v}_0 = 3\mathbf{v}_0. \tag{5}$$

(b) Since maximum height scales as $\mathbf{v}^2/(2\mathbf{g})$, tripling the launch speed multiplies the height by $3^2 = \mathbf{9}$, so the tennis ball reaches $\mathbf{h}_{\text{max}} = \mathbf{9h}$ (drawing the extra kinetic energy from a tiny slowdown of the heavy basketball). Spacecrafts perform the same trick gravitationally: swinging behind an orbiting planet is an elastic scatter in the planet's rest frame, adding up to twice the planet's orbital speed in the Sun's frame without burning fuel!

Problem 4.

(a) Because the upper portion of the chain is in free fall (no tension acts between loose adjacent links), after descending distance $\mathbf{y}$ every falling link has downward speed $\mathbf{v} = \sqrt{(2\mathbf{gy})}$. In a short time $\mathbf{dt}$, a segment of length $\mathbf{dy} = \mathbf{v}\ \mathbf{dt}$ and linear mass density $\lambda = \mathbf{M}/\mathbf{L}$ reaches the pan, so mass arrives at rate

$$(\mathbf{dm})/\ (\mathbf{dt}) = \lambda\mathbf{v} = (\mathbf{M})/\ (\mathbf{L})\ \sqrt{(2\mathbf{gy})}. \tag{6}$$

(b) Two effects contribute to the normal force $\mathbf{F}$ exerted by the scale: (1) supporting the weight $(\lambda\mathbf{y})\mathbf{g}$ of the length $\mathbf{y}$ already resting on the pan, and (2) supplying the upward impulse per unit time needed to decelerate incoming mass $\mathbf{dm}/\mathbf{dt}$ from speed $\mathbf{v}$ to 0, namely $\mathbf{F}_{\text{imp}} = \mathbf{v}(\mathbf{dm}/\mathbf{dt}) = \lambda\mathbf{v}^2 = \lambda(2\mathbf{gy})$. Adding both contributions yields

$$\mathbf{F}(\mathbf{y}) = \lambda\mathbf{gy} + 2\lambda\mathbf{gy} = 3\lambda\mathbf{gy} = (3\mathbf{Mg})/\ (\mathbf{L})\ \mathbf{y}. \tag{7}$$

Just before the top link lands ($\mathbf{y} \to \mathbf{L}^-$), the scale reads $\mathbf{3Mg}$—three times the chain's weight! Once the entire chain has come to rest on the pan, the impact term vanishes and the reading drops abruptly to $\mathbf{Mg}$.